MEHEDI HASAN

Python, Automation & Full-Stack Tooling Developer | Undergraduate CSE Researcher

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PROFESSIONAL & RESEARCH EXPERIENCE

FEC CSE Research Lab, Faridpur Engineering College **Present (Active)**

Undergraduate Research Assistant *Faridpur, Bangladesh*

- Conducting applied Computer Science & Engineering research in collaboration with faculty advisors and lab peers, focusing on empirical evaluation, verification pipelines, and reproducible systems.
- Designing experimental benchmarks, data processing workflows, and preparing technical manuscripts for upcoming peer-reviewed academic publication.

Independent Full-Stack & Automation Systems Developer (@MEhe132) **Present (Active)**

Python, Full-Stack Web & OS-Level Tooling Engineer *Remote / Dhaka, Bangladesh*

- Architected and deployed full-stack web applications on Vercel (Django, Flask, SQLAlchemy, Leaflet.js GIS) serving real-time civic tracking and multi-user financial auditing.
- Engineered OS-level Windows desktop utilities using Python **ctypes** (Win32 API) and published open-source Manifest V3 browser automation extensions for Microsoft Edge and Google Chrome.

EDUCATION

Faridpur Engineering College (FEC) **Present (Undergraduate)**

Bachelor of Science (B.Sc.) in Computer Science & Engineering (CSE) *Faridpur, Bangladesh*

- **Relevant Coursework & Focus:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, and Software Engineering.
- **Research Affiliation:** Active member and Undergraduate Research Assistant at the **FEC CSE Research Lab**.

SELECTED ENGINEERING PROJECTS (REVERSE-CHRONOLOGICAL)

Fix_my_city — Bangladesh Civic Infrastructure Issue Tracker & GIS Platform **2026 – Present**

Python, Django, Leaflet.js (GIS), ReportLab (PDF), EXIF/AI Forensic Analysis *Live Demo | GitHub*

- Built a two-tier civic-tech platform (Citizen Reporter vs. Municipal Authority Triage Console) with immutable audit trails (**StatusHistory**) across Dhaka municipal wards (DNCC & DSCC).
- Engineered a hybrid AI-generated image & spam verification pipeline combining Camera Hardware EXIF fast-path checks with binary header/noise detection blocking Midjourney/DALL-E uploads.
- Automated statutory PDF petition generation under the Local Government Act 2009 & RTI Act 2009 using **ReportLab** with GPS watermarks and multi-layer **Leaflet.js** Nominatim reverse geocoding.

Meal_management_web_app (Plate & Spoon) — Real-Time Hostel Meal & Auditing System **2026 – Present**

Python, Flask, SQLAlchemy ORM, Server-Sent Events (SSE), PostgreSQL/SQLite, Tailwind CSS *Live App | GitHub*

- Implemented a thread-safe real-time **Server-Sent Events (SSE)** streaming engine (**MessageAnnouncer** using **threading.Lock()**) for non-blocking live group chat and instant financial notifications.
- Reduced dashboard query complexity from 100+ N+1 queries to 3–4 sub-millisecond queries via composite SQLite/PostgreSQL indexes and bulk aggregation (**get_bulk_user_monthly_stats**).

Convert_php_to_usdt (calc-php-converter v4.0.0) — Win32 Desktop OCR & Currency Overlay **2026 (Release v4.0.0)**

Python, Tkinter GUI, Windows Win32 API (ctypes: User32/Shcore), Pillow (PIL) OCR *Release v4.0.0 | GitHub*

- Developed a frameless, always-on-top Windows overlay utility with direct Win32 API integration for global hotkeys (**Ctrl+/** to summon, **Ctrl++/-** for live opacity) and High-DPI scaling.
- Integrated a Google Lens OCR screen-region snipping tool, dynamic UI scaling buttons, and persistent session state (**window_config.json**).

Safe-Seller-Keep-Alive (v1.3.0) — Stealth Marketplace Session Manager Extension **2026 (Release v1.3.0)**

JavaScript (ES6+), Chrome & Edge Extension Manifest V3 (chrome.alarms, chrome.power) *Release v1.3.0 | GitHub*

- Built a silent Manifest V3 background extension maintaining active G2G & U7BUY seller sessions using randomized jitter intervals (600s–1000s) without stealing window focus.
- Added Windows S0 Modern Standby wake-lock protection (**chrome.power**), automatic countdown recovery after PC restarts, and Cloudflare challenge / logout auto-pause safeguards.

TECHNICAL SKILLS

Languages:	Python (Primary), C, C++, Java, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
Frameworks & Libraries:	Django, Flask, SQLAlchemy ORM, Tkinter, Tailwind CSS, Node.js, Leaflet.js (GIS), ReportLab, Pillow (PIL)
Systems & Tooling:	Windows Win32 API (ctypes), Chrome/Edge Extension APIs (Manifest V3), Server-Sent Events (SSE), Git, GitHub Actions, Vercel, Linux